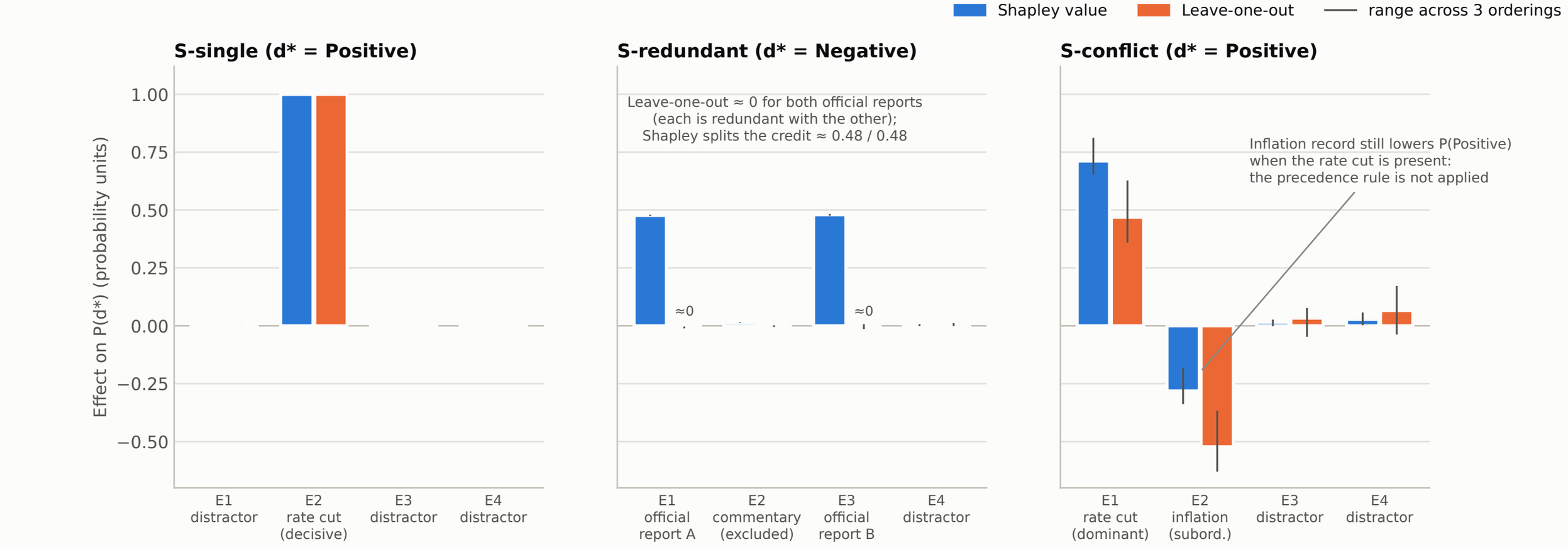
Synthetic controls: Shapley recovers single and redundant causes that leave-one-out can miss
Shapley value
Leave-one-out
range across 3 orderings
S-single (d* = Positive)
S-redundant (d* = Negative)
S-conflict (d* = Positive)
Effect on P(d*) (probability units)
1.00
0.75
0.50
0.25
0.00
−0.25
−0.50
Leave-one-out ≈ 0 for both official reports
(each is redundant with the other);
Shapley splits the credit ≈ 0.48 / 0.48
≈0
≈0
Inflation record still lowers P(Positive)
when the rate cut is present:
the precedence rule is not applied
E1 distractor
E2 rate cut (decisive)
E3 distractor
E4 distractor
E1 official report A
E2 commentary (excluded)
E3 official report B
E4 distractor
E1 rate cut (dominant)
E2 inflation (subord.)
E3 distractor
E4 distractor